Basic Information

Course: EE660: Mathematical Foundations and Methods of Machine Learning
Term: Fall 2026
Instructor: Stephen Tu (stephen.tu@usc.edu)
TAs: Yichen Zhou (yzhou222@usc.edu)

Please include [EE660] in the subject of any email related to the course.

Time and Location:

Lecture
Tuesday and Thursday, 4:00-5:50pm, OHE 100D.

Discussion
Friday, 2:00-2:50pm, OHE 100D.

Midterm:
10/6, Tuesday, 4:00-5:50pm, Location OHE 100D.

Final:
12/10, Thursday, 4:30-6:30pm, Location **TBD**.

Office Hours:
Stephen: Thursday, 3:00-4:00pm, EEB 326 and Zoom (Hybrid)
Zoom link: https://usc.zoom.us/j/99442461937?pwd=vEthj4uKR8s89BoFzzFqZtR7Wf1aru.1

Yichen: Friday, 3:00-4:00pm, PHE 410 and Zoom (Hybrid)
Zoom link: https://usc.zoom.us/j/98459459440

Course Prereqs: EE 503, EE 510, and EE 559

Course Description

This course will cover the mathematical foundations of machine learning, including both supervised/unsupervised learning. The focus will be on the mathematical principles underlying machine learning algorithms. Topics covered include (but are not limited to) Perceptron, support vector machines, kernel methods, random features, concentration inequalities, empirical process theory, VC-dimension, stochastic gradient methods, nearest neighbors classification, universal approximation for neural networks, principle component analysis, maximum likelihood estimation, latent variable models, expectation-maximization, energy-based models, score matching, variational autoencoders, normalizing flows. Students will leave this course with a rigorous understanding of classic machine learning methods.

Textbook

There is no textbook for the course. Instead the reading will be from the course lecture notes.

Note: I will frequently update the lecture notes as the course progresses (filling in new content as we move forward in addition to correcting typos). While I will update the DEN with newer versions of the notes, I recommend instead using the link (https://stephentu.github.io/pdfs/EE660_Lecture_Notes.pdf) and refreshing each time you go to read.

Furthermore, as these notes are a work in progress, please feel free to send me an email if you suspect any typos or errors.

Homeworks (Optional)

There will be 5 **optional** homeworks assignments throughout the semester. These homeworks will mostly be pen and paper exercises, with an occasional coding exercise. For any coding exercises, you are free to use whatever language you choose (although we highly encourage the use of Python and its comprehensive scientific computing ecosystem). As the homework assignments are optional, they do **not** need to be turned in for grading.

In-Class Quizzes

There will be 5 in-class quizzes, which will be held for 15 minutes at the beginning of class. The in-class quizzes will be closed-book, and will consist of exactly one problem taken verbatim from the corresponding homework assignment. DEN students will use the Honorlock remote proctoring tool for taking the in-class quizzes.

Grading Breakdown

The following table describes how your final grade will be calculated.

Assignment	**Weight**
In-Class Quizzes	30%
Midterm	30%
Final	40%

Grading notes: You **must** take and turn in both the midterm and final to pass the class.

For the in-class quizzes, each of the five quizzes will count uniformly towards your grade.

AI Chatbot Policy

USC's official stance, as outlined by the Academic Senate, is that each course may determine its own policy on the acceptable use of AI assistants. Accordingly, we outline EE660's policy below. For simplicity, we use ChatGPT as a generic term to refer to AI chatbots such as ChatGPT, Gemini, Claude, Grok, and others.

In this course, you are free and encouraged to use ChatGPT in any way you see fit for learning the material. This can include, but is certainly not limited to, asking ChatGPT to explain parts of the lecture notes, give hints and/or solutions to the homework problems, and generate practice problems for the midterm/final. Of course, ChatGPT is not permitted to be used in any in-class/remote proctored exams, so while ChatGPT is an invaluable learning aid, students are encouraged to remember that it is not intended to replace the learning process.

If you have any specific questions about valid ChatGPT usage, please ask the course staff.

Topics

Supervised Learning

- Perceptron
  - Mistake bound.
  - Generalization bound.
- Support Vector Machines / Kernel Machines
  - SVM primal/dual formulation.
  - Kernels from both SVM and least-squares duality.
  - Positive definite kernels.
  - Random features.
  - Gaussian processes.
- Empirical Risk Minimization (short introduction to empirical process theory)
  - Basic concentration inequalities.
  - Loss functions and margin theory.
  - Uniform convergence and Rademacher complexity.
  - VC-dimension and the no free lunch theorem.
- First order / stochastic gradient methods
  - Basic framework and convergence results.

Unsupervised Learning and Probabilistic Modeling

- Dimensionality reduction
    - Random projections.
    - Principle components analysis.
- K-means clustering
- Maximum likelihood estimation
- Latent variable models
    - Expectation-maximization.
    - Evidence lower bound (ELBO) and variational inference.
- Sequence modelling
    - N-gram models and MLE.
    - Hidden markov models.
    - Recurrent neural networks, state-space models.
    - Attention models.

Schedule

Note: While the dates of the In-Class Quizzes and Midterm exam are set, the exact set of topics covered during each lecture are subject to change depending on our pace during the semester.

The section(s) in the **Reading and HW** column refer to the lecture notes.

Date	**Topics**	**Reading and HW/Quizzes**
8/25	Intro, binary classification, Perceptron algorithm.	Sections 1.1.
8/27	Perceptron mistake bound.	Sections 1.2.1.
9/1	Perceptron risk bound, support vector machines (SVMs).	(HW1 released.) Sections 1.2.2, 1.3.1.
9/3	SVM duality.	Section 1.3.1.
9/8	Feature maps, kernels, universality.	Section 1.3.2.
9/10	Random features.	Section 1.3.3.
9/15	Gaussian processes.	**HW1 in-class quiz.**
9/17	Empirical risk minimization (ERM).	(HW2 released.) Sections 1.4.1 to 1.4.3.
9/22	Generalization for finite classes, Rademacher complexity and	Section 1.4.4, 1.4.7.

	symmetrization.	
9/24	Explicit computations of Rademacher complexity.	Section 1.4.9.
9/29	Sub-Gaussian random variables, basic concentration inequalities.	Section B.
10/1	Generalization bounds for SVMs.	**HW2 in-class quiz.**
10/6	**Midterm**	
10/8	**Holiday: Fall Recess**	
10/13	VC dimension and examples.	(HW3 released.) Section 1.4.10.
10/15	Finish VC dimension, PAC learnability, and the no free lunch theorem.	Sections 1.4.11, 1.4.12.
10/20	First-order and stochastic optimization.	Section 1.5.
10/22	Stochastic optimization continued.	
10/27	Dimensionality reduction: random projections and PCA.	**HW3 in-class quiz.** Section 2.1.
10/29	K-means clustering.	(HW4 released.) Section 2.2.
11/3	Maximum likelihood estimation.	Section 2.3.
11/5	Latent variable models, expectation maximization (EM).	Section 2.7.
11/10	Variational inference, ELBO.	Section 2.8.
11/12	Sequence modelling, N-gram models.	**HW4 in-class quiz.** (HW5 released.)
11/17	Hidden markov models, inference.	
11/19	HMM estimation.	
11/24	Recurrent neural networks, state-space models.	
11/26	**Holiday: Thanksgiving Break**	

12/1	Attention models.	**HW5 in-class quiz.**
12/3	Conclusion.	